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$$\begin{aligned} I &= \int \frac{\ln^2 x}{x} dx \\ \begin{cases} u = \ln^2 x \\ dv = \frac{1}{x} dx \end{cases} &\Rightarrow \begin{cases} du = 2 \frac{\ln x}{x} \\ v = \ln x \end{cases} \\ I &= \ln x \cdot \ln^2 x - 2 \int \frac{\ln^2 x}{x} dx \\ I &= \ln x \cdot \ln^2 x - 2I + c \\ 3I &= \ln x \cdot \ln^2 x + c \\ I &= \frac{\ln x \cdot \ln^2 x}{3} + c \end{aligned}$$

References